UNTRM AI WORKSHOP · SESSION 2

Inside the Black Box: Foundations of Deep Learning

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Chachapoyas · Monday, November 9, 2026

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FROM SESSION 1

Where we left off

Everything is numbers

Images, text and DNA become tensors with a known shape

Deep learning learns its features

Nobody tells it what to look for in the data

You trained a classifier

Bean leaf photos in Colab, in a few minutes

Your bean classifier got most leaves right. What did it actually learn?

This session's plan

PART 1 · 25 MIN

Why deep learning?

How models learn

PART 2 · 20 MIN

Convolutional networks

Images and DNA

PART 3 · 20 MIN

Recurrent networks

Time series

PART 4 · 15 MIN

Autoencoders

Anomalies, denoising

PART 5 · 25 MIN

Transformers

Attention, proteins, DNA

WRAP-UP · 10 MIN

Which architecture?

Match model to data

PART 1

Why deep learning?

How computers learn from examples, how we know a model generalizes, and when deep learning beats classical methods

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SHOW OF HANDS

How computers learn: guess the rule

Examples A

[1, 1] → 2

[2, 1] → 3

[2, 2] → 4

[3, 2] → ?

Examples B

[1, 1] → 3

[2, 1] → 5

[2, 2] → 6

[3, 2] → ?

Write down your answers before the next slide.

HOW COMPUTERS LEARN

Two rules fit Examples B perfectly

RULE 1

$2x + y$

8

RULE 2

$x^2 - x + y + 2$

10

Both give 3, 5 and 6 for the three examples. They only disagree on the new input [3, 2].

Generalization

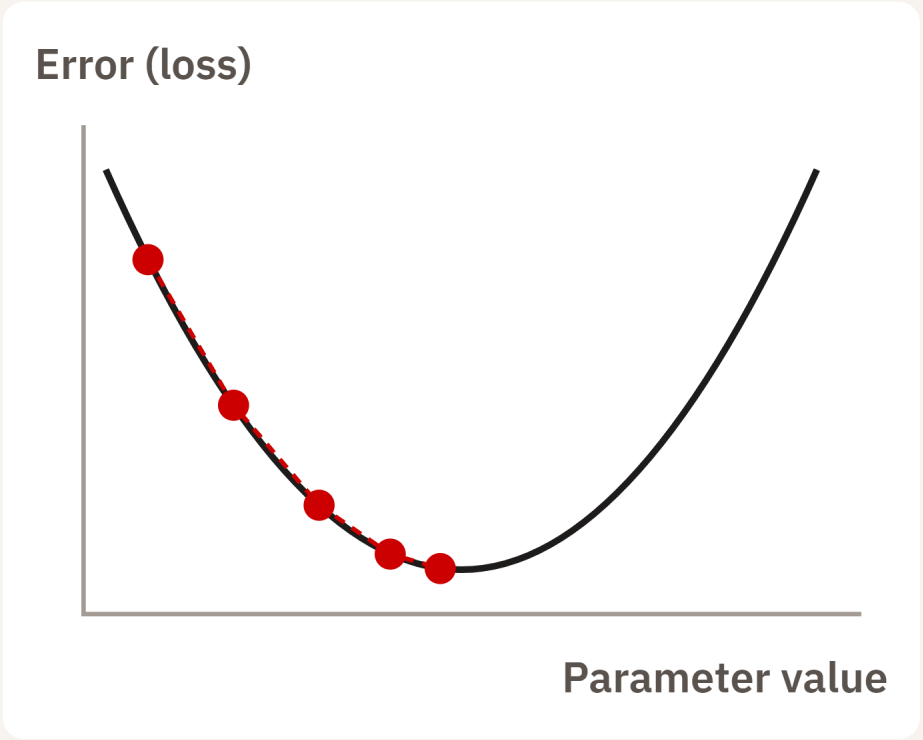
Test a model on data it has never seen

Model fit

Match the model's flexibility to the data you have

FITTING A MODEL

Training means walking downhill on the error



Parameters

The knobs the model can turn

Loss

One number for how wrong the predictions are

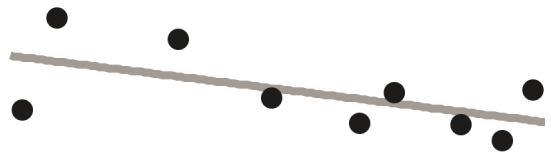
Gradient descent

Turn every knob a little in the direction that lowers the loss, and repeat

Capacity: how flexible the model is, like a polynomial's degree or a network's size.

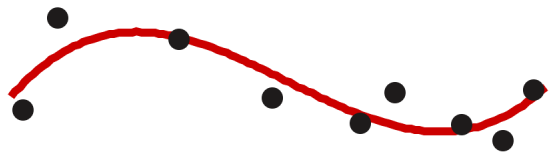
UNDERFITTING VS. OVERFITTING

Too simple, just right, too flexible



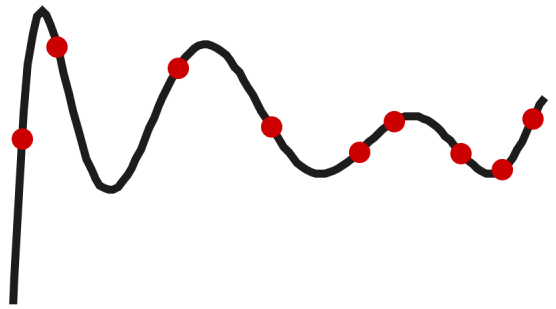
Underfit

Too rigid: misses the pattern



Good fit

Follows the trend, not the noise



Overfit

Too flexible: chases every point

The goal is low error on new data, not a perfect fit to the training data.

HOW WE KNOW IT WORKED

Train, validate, then test once



Little data?

Cross-validation: rotate which part is held out, then average the scores

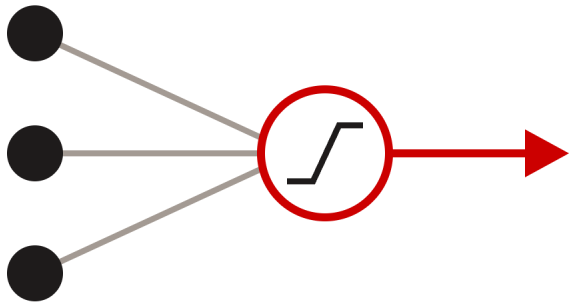
Field and lab data

Hold out whole plots, sites, seasons or slides, not random rows

FROM NEURON TO NETWORK

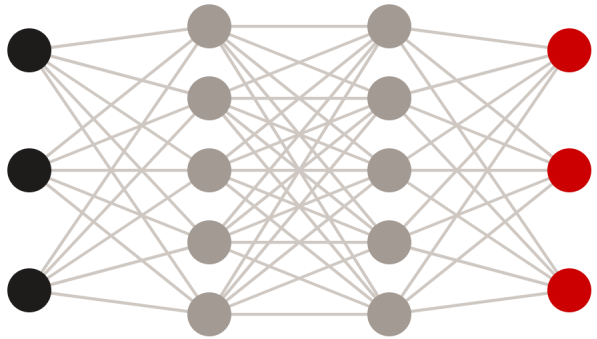
A network is many simple units, stacked

One neuron



Multiply each input by a **weight**, add them up, then apply a simple **nonlinearity**

Layers of neurons (MLP)



Hidden layers **build features**; the last layer **decides** (yield, class, risk)

LIVE DEMO · 3 MIN

molab.marimo.io · Classification MLP

Watch a network learn its own features

1

Three classes that no straight line can separate

2

Train: watch the learning curve go down

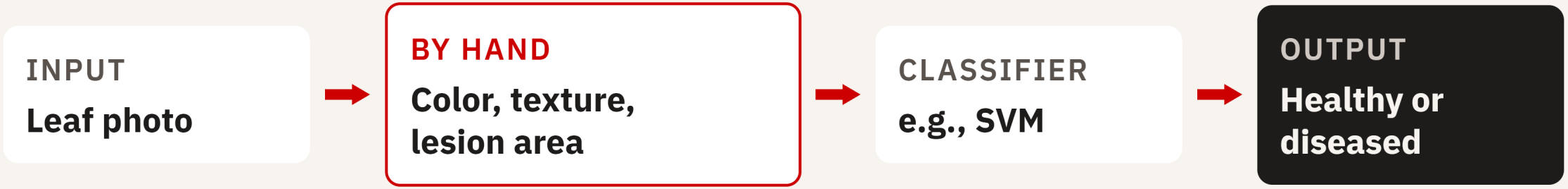
3

Look inside the last hidden layer: the classes now sit apart

That last picture is what “the network learns its own features” means.

CLASSICAL ML VS. DEEP LEARNING

When hand-built features hit their limit



It works for the crop, camera and lighting it was tuned on. A new crop, a new camera or a cloudy day often means designing the features again.

The bottleneck is not the classifier. It is everything done by hand before it.

CLASSICAL ML VS. DEEP LEARNING

Four places classical ML hits a wall

Raw, high-dimensional data

Full DNA sequences or hyperspectral pixels are too big to use directly

Features built by hand

Spore or nematode shape features redesigned for every new species

Layered structure

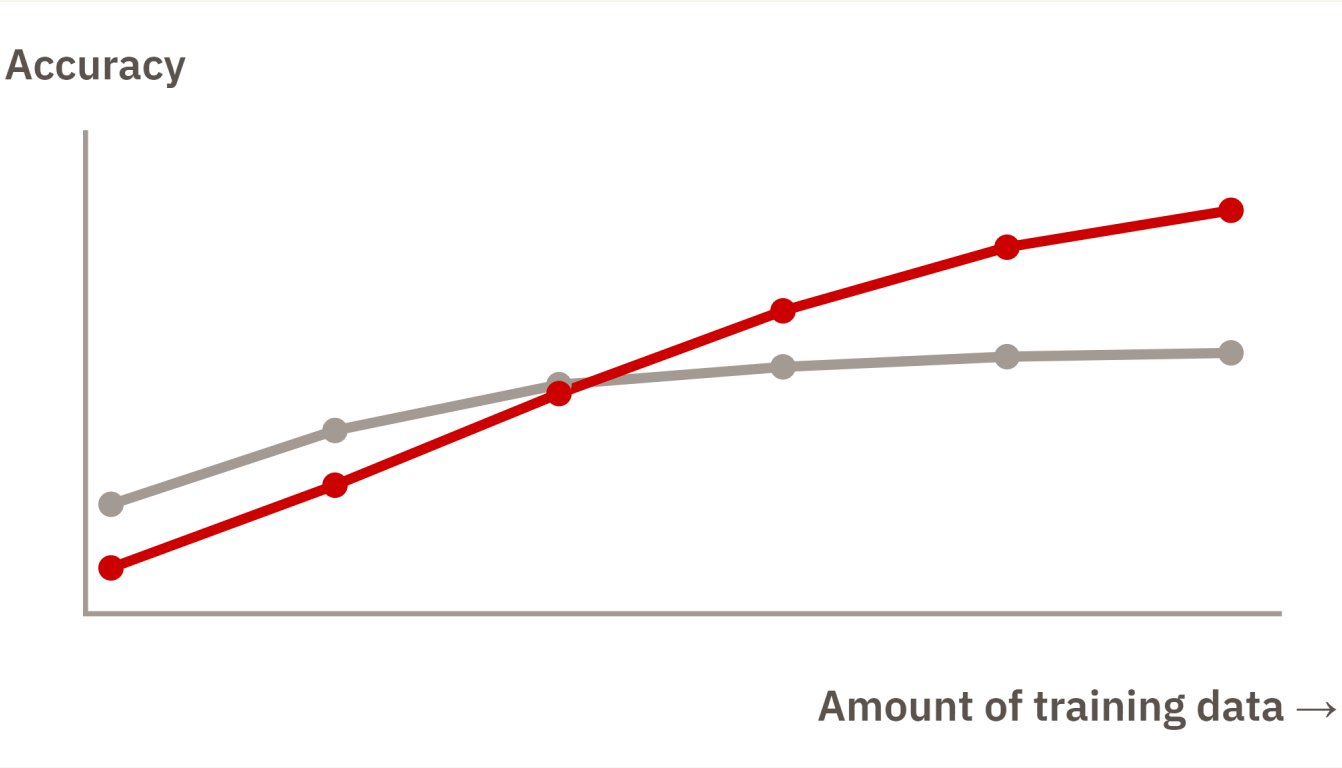
Drone imagery: pixels → leaves → plants → damage across a field

Combining data types

Drone images + soil tests + weather need hand-made rules to fuse

THE PERFORMANCE PLATEAU

More data keeps helping deep learning for longer



Deep learning
Learned features keep improving

Classical ML
Hand-built features level off

With little data, classical methods often win. The lines cross only once you have enough examples.

Illustrative curves

CLASSICAL ML VS. DEEP LEARNING

What deep learning costs you

Needs a lot of data
Rare pest outbreaks give few labeled examples

Needs compute
Training uses GPUs; Tuesday we use the university HPC

Hard to interpret
An agronomist must be able to justify a treatment

Fragile
New light, camera or sensor noise can flip a prediction

Neither approach is strictly better.

Classical ML wins when

data are scarce, the problem is simple, or you must explain the result

Deep learning earns its cost on

images, sequences, and several data types combined

Next: what deep networks look like inside, starting with images.